By *architecture* as a *process*, we mean an act of translating architectural drivers into the architecture design and architectural decisions that shape the structure.

What do we *not* consider architecture? Details of the code, or how the particular line of code was written. Everything that can be easily changed within minutes is usually not an architectural concern.

In this chapter, we focus on both sides: architecture as a structure of related building blocks, and architecture as a process of designing and making conscious decisions.

9.1.2 How to document architecture: The C4 model

One great author who strongly influences our understanding of software architecture is Simon Brown. In *C4 Model for Visualising Software Architecture* (Leanpub, 2022), he shares a brief yet powerful notation that can be used to capture architecture diagrams: the C4 model.

We describe this notation here for two reasons. First, to explain to you ways of designing data products in our Messflix example. Second, it is common that in big organizations, architecture of every project is documented differently, which makes an exchange of knowledge and cross-reviews problematic. By using the C4 notation, you can introduce a common language to talk about architecture in your organization.

The C4 model is built with four levels of abstraction:

1. *Context*—It shows a software system as a black box with people and other systems interacting with it.
2. *Container*—It is similar to a context diagram, but reveals the containers that form the software system. By *containers*, Brown usually means applications or data stores. In the container diagram, we also extend that definition with descriptions of used technologies (like Java with Spring Boot or Oracle Database) and communication mechanisms (REST call or Kafka message).